Urban Risk Analytics API (0.1.0)

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tiles

This is an Endpoint to get the road tiles in MVT format with Risk Overlays

Returns a Mapbox Vector Tile (MVT) for roads in the requested slippy-map tile. Risk scoring is optional and controlled by the `crime`, `collisions`, and `userReportedEvents` toggles. When any toggle is enabled, both `startMonth` and `endMonth` are required in `YYYY-MM` format.

PATH PARAMETERS

`z` required	integer (Z) [0 .. 22] Slippy-map zoom level (0-22).
`x` required	integer (X) >= 0 Slippy-map tile X coordinate.
`y` required	integer (Y) >= 0 Slippy-map tile Y coordinate.

QUERY PARAMETERS

`startMonth` >	Startmonth (string) or Startmonth (null) (Startmonth) Inclusive start month for risk scoring in `YYYY-MM` format. Required when any risk toggle is true.
`endMonth` >	Endmonth (string) or Endmonth (null) (Endmonth) Inclusive end month for risk scoring in `YYYY-MM` format. Required when any risk toggle is true.
`crime` >	Crime (boolean) or Crime (null) (Crime) Include crime component in risk scoring.

`collisions` >	Collisions (boolean) or Collisions (null) (Collisions) Include collisions component in risk scoring.
`userReportedEvents` >	Userreportedevents (boolean) or Userreportedevents (null) (Userreportedevents) Include user-reported events component in risk scoring.

Responses

> **200** Binary Mapbox Vector Tile payload (`application/vnd.mapbox-vector-tile`).

> **400**
Invalid request. Examples: removed `month` or `crimeType`, invalid tile coordinates, invalid month format, or missing month window when risk toggles are enabled.

> **422** Validation Error

> **503** Database dependency unavailable while generating tile.

GET `/tiles/roads/{z}/{x}/{y}.mvt`

Response samples

400 | 422 | 503

Content type
application/json

Example
Removed month parameter

Copy Expand all Collapse all

```
{
    "error": "MONTH_PARAMETER_REMOVED",
    "message": "month has been removed; use startMonth and endMonth instead",
  - "details": {
        "field": "month"
    }
}
```

auth

Register a new user

Creates a user account with email and password.

REQUEST BODY SCHEMA: application/json

`email` required	string (Email) Email address for the account.
`password` required	string (Password) >= 8 characters Plain-text password. Must be at least 8 characters.

Responses

> **200** Registered user profile.

> **422** Validation Error

POST `/auth/register`

Request samples

Payload

Content type
application/json

Copy

```
{
    "email": "student@example.com",
    "password": "StrongPass123"
}
```

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
 - "user": {
      "id": 0,
      "email": "string",
      "is_admin": true,
      "created_at": "2019-08-24T14:15:22Z"
   }
}
```


Log in with email and password

Authenticates a user and returns a bearer access token.

REQUEST BODY SCHEMA: application/json

email required	string (Email) Email address for the account.
password required	string (Password) >= 8 characters Plain-text password. Must be at least 8 characters.

Responses

> 200 Access token and authenticated user profile.

> 422 Validation Error

POST /auth/login

Request samples

Payload

Content type
application/json

Copy

```
{
    "email": "student@example.com",
    "password": "StrongPass123"
}
```

Response samples

200 **422**

Content type
application/json

Copy Expand all Collapse all

```
{
    "access_token": "string",
    "token_type": "string",
  - "user": {
        "id": 0,
        "email": "string",
        "is_admin": true,
        "created_at": "2019-08-24T14:15:22Z"
    }
}
```

Get current user

Returns the currently authenticated user from the bearer token.

AUTHORIZATIONS: > *HTTPBearer*

Responses

> **200** Current authenticated user profile.

GET /me

Response samples

200

Content type
application/json

Copy Expand all Collapse all


```
{
 - "user": {
      "id": 0,
      "email": "string",
      "is_admin": true,
      "created_at": "2019-08-24T14:15:22Z"
   }
}
```

Update current user profile

Updates the authenticated user's email and/or password.

AUTHORIZATIONS: > *HTTPBearer*

REQUEST BODY SCHEMA: application/json

`email` >	Email (string) or Email (null) (Email) New email address. Optional.
`password` >	Password (string) or Password (null) (Password) New plain-text password. Optional, minimum 8 characters.

Responses

> **200** Updated user profile.

> **422** Validation Error

PATCH /me

Request samples

Payload

Content type
application/json

Copy

```
{
    "email": "new-email@example.com"
}
```

Response samples

200 **422**

Content type
application/json

Copy Expand all Collapse all

```
{
  - "user": {
        "id": 0,
        "email": "string",
        "is_admin": true,
        "created_at": "2019-08-24T14:15:22Z"
    }
}
```

---

reported-events

Create a reported event

Creates a new user-reported event (crime or collision). If a bearer token is provided, the report is linked to the authenticated user; otherwise it is stored as anonymous.

AUTHORIZATIONS: > *HTTPBearer*

REQUEST BODY SCHEMA: application/json

`event_kind` required	string (Event Kind) Enum: `"crime"` `"collision"` Type of reported event.
`event_date` required	string <date> (Event Date) Calendar date when the event happened (`YYYY-MM-DD`).
`event_time` >	Event Time (string) or Event Time (null) (Event Time) Optional local time when the event happened (`HH:MM`).
`longitude` required	number (Longitude) [-180 .. 180] Event longitude in WGS84.
`latitude` required	number (Latitude) [-90 .. 90] Event latitude in WGS84.
`description` >	Description (string) or Description (null) (Description) Optional free-text description from the reporter.
`crime` >	ReportedCrimePayload (object) or null Required when `event_kind=crime` ; must be omitted for collision events.
`collision` >	ReportedCollisionPayload (object) or null Required when `event_kind=collision` ; must be omitted for crime events.

Responses

> **201** Reported event created successfully.

— **400** Invalid request payload or validation error.

— **401** Authentication token provided but invalid/expired.

> **422** Validation Error

— **503** Database unavailable.

POST /reported-events

Request samples

Payload

Content type
application/json

Copy Expand all Collapse all


```
{
 - "crime": {
      "crime_type": "Burglary"
   },
   "description": "Suspicious activity near junction after 9pm.",
   "event_date": "2026-03-10",
   "event_kind": "crime",
   "event_time": "21:30",
   "latitude": 53.699,
   "longitude": -1.518
}
```


Response samples

201 **422**

Content type
application/json

Copy Expand all Collapse all

```
{
 - "report": {
      "id": 1024,
      "event_kind": "crime",
      "reporter_type": "authenticated",
      "month": "2026-03",
      "event_date": "2026-03-10",
      "event_time": "21:30",
      "longitude": -1.518,
      "latitude": 53.699,
      "segment_id": 0,
      "snap_distance_m": 0,
      "description": "string",
      "admin_approved": true,
      "moderation_status": "string",
      "moderation_notes": "string",
      "created_at": "string",
    + "details": { … },
      "user_id": 0,
      "reporter_email": "string",
      "moderated_by": 0,
      "moderated_at": "string"
   }
}
```

List my reported events

Returns a cursor-paginated list of reports submitted by the currently authenticated user. Supports optional moderation status and event-kind filters.

AUTHORIZATIONS: > *HTTPBearer*

QUERY PARAMETERS

`limit`	integer (Limit) [1 .. 100] Default: `20` Maximum number of items to return per page.
`cursor` >	Cursor (string) or Cursor (null) (Cursor) Pagination cursor in `created_at|id` format.
`status` >	Status (string) or Status (null) (Status) Filter by moderation status: `pending`, `approved`, or `rejected`.

`event_kind` ›

Event Kind (string) or Event Kind (null) (Event Kind)
Filter by event kind: `crime` or `collision`.

Responses

› **200** My reports returned successfully.

— **400** Invalid filter/cursor parameters.

— **401** Authentication required.

› **422** Validation Error

— **503** Database unavailable.

GET /reported-events/mine

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
 - "items": [
     + { … }
   ],
 - "meta": {
       "returned": 0,
       "limit": 0,
       "nextCursor": "string",
       "filters": { }
   }
}
```

List reported events for admin moderation

Admin-only endpoint that returns a cursor-paginated report list with moderation, reporter identity, event kind, and month-range filtering.

AUTHORIZATIONS: > *HTTPBearer*

QUERY PARAMETERS

`from` >
From (string) or From (null) (From)
Inclusive start month in `YYYY-MM` format.

`to` >
To (string) or To (null) (To)
Inclusive end month in `YYYY-MM` format.

`limit`
integer (Limit) [1 .. 200]
Default: `50`
Maximum number of items to return per page.

`cursor` >
Cursor (string) or Cursor (null) (Cursor)
Pagination cursor in `created_at|id` format.

`status` >
Status (string) or Status (null) (Status)
Filter by moderation status: `pending`, `approved`, or `rejected`.

`event_kind` >
Event Kind (string) or Event Kind (null) (Event Kind)
Filter by event kind: `crime` or `collision`.

`reporter_type` >
Reporter Type (string) or Reporter Type (null) (Reporter Type)
Filter by reporter identity type: `anonymous` or `authenticated`.

Responses

> **200** Admin report list returned successfully.

— **400** Invalid filter/month/cursor parameters.

— **401** Authentication required.

— **403** Admin access required.

> **422** Validation Error

— **503** Database unavailable.

GET /admin/reported-events

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
 - "items": [
     + { … }
   ],
 - "meta": {
       "returned": 0,
       "limit": 0,
       "nextCursor": "string",
       "filters": { }
   }
}
```


Moderate a reported event

Admin-only endpoint to approve or reject a reported event and optionally add moderation notes.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

`report_id` required	integer (Report Id) Unique ID of the reported event to moderate.

REQUEST BODY SCHEMA: application/json

`moderation_status` required	string (Moderation Status) Enum: `"approved"` `"rejected"` Admin moderation decision for the report.

`moderation_notes` > Moderation Notes (string) or Moderation Notes (null) (Moderation Notes)
Optional admin notes explaining the moderation decision.

Responses

> **200** Report moderation applied successfully.

— **400** Invalid moderation payload.

— **401** Authentication required.

— **403** Admin access required.

— **404** Reported event not found.

> **422** Validation Error

— **503** Database unavailable.

PATCH `/admin/reported-events/{report_id}/moderation`

Request samples

Payload

Content type
application/json

Copy

```
{
    "moderation_notes": "Corroborated with local incident cluster.",
    "moderation_status": "approved"
}
```

Response samples

200 **422**

Content type
application/json

Copy Expand all Collapse all


```
{
 - "report": {
      "id": 1024,
      "event_kind": "crime",
      "reporter_type": "authenticated",
      "month": "2026-03",
      "event_date": "2026-03-10",
      "event_time": "21:30",
      "longitude": -1.518,
      "latitude": 53.699,
      "segment_id": 0,
      "snap_distance_m": 0,
      "description": "string",
      "admin_approved": true,
      "moderation_status": "string",
      "moderation_notes": "string",
      "created_at": "string",
    + "details": { … },
      "user_id": 0,
      "reporter_email": "string",
      "moderated_by": 0,
      "moderated_at": "string"
   }
}
```


List user-reported events as GeoJSON

Returns user-reported events as a GeoJSON FeatureCollection for map display. Supports moderation, type, month-range, admin-approved, and optional bounding-box filters.

QUERY PARAMETERS

`from` >	From (string) or From (null) (From) Inclusive start month in `YYYY-MM` format.
`to` >	To (string) or To (null) (To) Inclusive end month in `YYYY-MM` format.
`adminApproved` >	Adminapproved (boolean) or Adminapproved (null) (Adminapproved) Filter by whether event was admin-approved.
`minLon` >	Minlon (number) or Minlon (null) (Minlon) Bounding box min longitude.

`minLat` >	Minlat (number) or Minlat (null) (Minlat) Bounding box min latitude.
`maxLon` >	Maxlon (number) or Maxlon (null) (Maxlon) Bounding box max longitude.
`maxLat` >	Maxlat (number) or Maxlat (null) (Maxlat) Bounding box max latitude.
`limit`	integer (Limit) [1 .. 5000] Default: `500` Maximum number of features to return.
`status` >	Status (string) or Status (null) (Status) Filter by moderation status: `pending`, `approved`, or `rejected`.
`event_kind` >	Event Kind (string) or Event Kind (null) (Event Kind) Filter by event kind: `crime` or `collision`.
`reporter_type` >	Reporter Type (string) or Reporter Type (null) (Reporter Type) Filter by reporter identity type: `anonymous` or `authenticated`.

Responses

> **200** GeoJSON user events returned successfully.

— **400** Invalid filter/month/bounding-box parameters.

> **422** Validation Error

— **503** Database unavailable.

GET `/user-events`

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all

```
{
    "type": "FeatureCollection",
  - "features": [
      + { … }
    ],
  - "meta": {
        "returned": 0,
        "limit": 0,
        "filters": { }
    }
}
```


watchlists

Read Watchlists

AUTHORIZATIONS: > *HTTPBearer*

QUERY PARAMETERS

`watchlist_id` > Watchlist Id (integer) or Watchlist Id (null) (Watchlist Id)

Responses

> **200** Successful Response

> **422** Validation Error

GET /watchlists

Response samples

200 422

Content type
application/json

Example
WatchlistListResponse

Copy Expand all Collapse all

```
{
 - "items": [
     + { … }
   ]
}
```

Create Watchlist

AUTHORIZATIONS: > *HTTPBearer*

REQUEST BODY SCHEMA: application/json

`name` required	string (Name) non-empty
`min_lon` required	number (Min Lon)
`min_lat` required	number (Min Lat)
`max_lon` required	number (Max Lon)
`max_lat` required	number (Max Lat)
`preference` >	WatchlistPreferencePayload (object) or null

Responses

> **201** Successful Response

> **422** Validation Error

POST /watchlists

Request samples

Payload

Content type
application/json

Copy Expand all Collapse all

```
{
    "name": "string",
    "min_lon": 0,
    "min_lat": 0,
    "max_lon": 0,
    "max_lat": 0,
  - "preference": {
        "start_month": "2019-08-24",
        "end_month": "2019-08-24",
      + "crime_types": [ … ],
        "travel_mode": "walk"
    }
}
```

Response samples

201 **422**

Content type
application/json

Copy Expand all Collapse all

```
{
 - "watchlist": {
      "id": 0,
      "user_id": 0,
      "name": "string",
      "min_lon": 0,
      "min_lat": 0,
      "max_lon": 0,
      "max_lat": 0,
      "created_at": "2019-08-24T14:15:22Z",
    + "preference": { … }
   }
}
```


Update Watchlist

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

`watchlist_id` required	integer (Watchlist Id)

REQUEST BODY SCHEMA: application/json

`name` >	Name (string) or Name (null) (Name)
`min_lon` >	Min Lon (number) or Min Lon (null) (Min Lon)
`min_lat` >	Min Lat (number) or Min Lat (null) (Min Lat)
`max_lon` >	Max Lon (number) or Max Lon (null) (Max Lon)
`max_lat` >	Max Lat (number) or Max Lat (null) (Max Lat)
`preference` >	WatchlistPreferencePayload (object) or null

Responses

> **200** Successful Response

> **422** Validation Error

PATCH /watchlists/{watchlist_id}

Request samples

Payload

Content type
application/json

Copy Expand all Collapse all


```
{
    "name": "string",
    "min_lon": 0,
    "min_lat": 0,
    "max_lon": 0,
    "max_lat": 0,
  - "preference": {
        "start_month": "2019-08-24",
        "end_month": "2019-08-24",
      + "crime_types": [ … ],
        "travel_mode": "walk"
    }
}
```


Response samples

200 **422**

Content type
application/json

Copy Expand all Collapse all

```
{
 - "watchlist": {
       "id": 0,
       "user_id": 0,
       "name": "string",
       "min_lon": 0,
       "min_lat": 0,
       "max_lon": 0,
       "max_lat": 0,
       "created_at": "2019-08-24T14:15:22Z",
     + "preference": { … }
   }
}
```


Delete Watchlist

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

`watchlist_id` required	integer (Watchlist Id)

Responses

> **200** Successful Response

> **422** Validation Error

DELETE `/watchlists/{watchlist_id}`

Response samples

200 | 422

Content type
application/json

Copy

```
{
    "deleted": true,
    "watchlist_id": 0
}
```


watchlist-analytics

Compute Watchlist Risk Score

Runs the watchlist risk algorithm using the watchlist bbox and stored preference window (start_month/end_month). The algorithm builds three core signals inside the bbox: (1) a crime component using crime-type harm weights, monthly recency decay, area normalization, and persistence across active months; (2) a collision component using severity-weighted collision points with recency decay normalized by effective road length; and (3) a lightweight user-report support signal with cluster caps and faster recency decay. Signals are blended with mode-specific weights (walk/drive) into a raw score, then mapped to a normalized 0-100 risk score. The run is persisted, then compared against prior runs with the same signature (month window + crime types + mode). If historical cohort size is below threshold, the service falls back to nearest reference bboxes and compares against cached or recalculated reference runs. Returns a condensed payload with risk result and comparison proof fields.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

`watchlist_id`
required

integer (Watchlist Id) > 0
Example: 42
Unique watchlist identifier owned by the authenticated user.

Responses

> **200** Risk score computed and persisted successfully.

— **400** Watchlist is missing required analytics preference fields or has invalid values.

— **404** Watchlist not found for the authenticated user.

> **422** Validation Error

POST /watchlists/{watchlist_id}/analytics/risk-score

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
    "watchlist_id": 42,
  - "risk_result": {
        "risk_score": 63,
        "raw_score": 1.6098,
    + "components": { … }
    },
  - "comparison": {
        "cohort_size": 2,
        "rank": 3,
        "rank_out_of": 2,
    + "reference_ids": [ … ]
    },
  - "data_used": {
        "official_crime_count": 937822,
        "collision_count": 714,
        "approved_user_report_count": 25
    },
  - "window": {
        "start_month": "2025-10-01",
        "end_month": "2026-01-01",
        "months_in_window": 4
    },
  - "normalization_context": {
        "area_km2": 12.41,
        "road_km": 185.22
    }
}
```

List Previous Watchlist Risk Runs

Returns previously persisted analytical risk-score runs for the selected watchlist, ordered newest first. Includes risk score, component breakdown, and stored comparison metadata for each historical run.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

`watchlist_id` required
integer (Watchlist Id) > 0
Example: `42`
Unique watchlist identifier owned by the authenticated user.

QUERY PARAMETERS

`limit`
integer (Limit) [1 .. 500]
Default: `50`
Example: `limit=50`
Maximum number of historical runs to return.

Responses

> **200** Historical risk runs returned successfully.

— **404** Watchlist not found for the authenticated user.

> **422** Validation Error

GET `/watchlists/{watchlist_id}/analytics/risk-score/runs`

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all

```
{
  "watchlist_id": 42,
```

```
  - "items": [
      + { … }
    ]
}
```

Get Basic Watchlist Metrics

Returns a lightweight analytics summary for the watchlist's stored bbox and month window (start_month to end_month). This endpoint is intentionally condensed and returns exactly 5 top-level outputs: number_of_crimes, number_of_collisions, number_of_user_reported_events, most_dangerous_roads (top 5), and crime_category_breakdown.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

`watchlist_id` required	integer (Watchlist Id) > 0 Example: 42 Unique watchlist identifier owned by the authenticated user.

Responses

> **200** Basic metrics returned successfully.

— **400** Watchlist has missing/invalid analytics month window or bbox values.

— **404** Watchlist not found for the authenticated user.

> **422** Validation Error

GET `/watchlists/{watchlist_id}/analytics/basic-metrics`

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
 - "crime_category_breakdown": [
    + { … },
    + { … }
   ],
 - "most_dangerous_roads": [
    + { … }
   ],
   "number_of_collisions": 27,
   "number_of_crimes": 1834,
   "number_of_user_reported_events": 11
}
```

Get Watchlist Map Events

Returns GeoJSON point layers scoped to the watchlist bbox and stored month window (start_month to end_month). The payload includes three FeatureCollections: official crimes, official collisions, and approved user-reported events. Designed for frontend map overlays alongside road tiles.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

`watchlist_id` required	integer (Watchlist Id) > 0 Example: 42 Unique watchlist identifier owned by the authenticated user.

Responses

> **200** Watchlist map event layers returned successfully.

— **400** Watchlist has missing/invalid month window or bbox values.

— **404** Watchlist not found for the authenticated user.

> **422** Validation Error

GET /watchlists/{watchlist_id}/analytics/map-events

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
 - "collisions": {
      "features": [ ],
      "type": "FeatureCollection"
   },
 - "crimes": {
    + "features": [ … ],
      "type": "FeatureCollection"
   },
 - "user_reported_events": {
      "features": [ ],
      "type": "FeatureCollection"
   }
}
```


Forecast Watchlist Next Month

Computes next-month forecast for a watchlist using the watchlist bbox plus historical monthly baseline from startMonth to the last complete month. The forecast blends official crime counts, approved user-reported crime signal, and collision severity points, then applies recency weights (lambda=0.85) and mode-based emphasis (walk/drive). Returns score, conservative band, expected crime/collision outputs, and Poisson-style intervals.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

`watchlist_id` required	integer (Watchlist Id) > 0 Example: 2 Unique watchlist identifier owned by the authenticated user.

REQUEST BODY SCHEMA: application/json

`startMonth` required	string (Startmonth) Historical baseline start month in YYYY-MM format.
`mode`	string (Mode) Default: `"walk"` Travel mode emphasis. Supported values: walk or drive (aliases accepted).
`crimeTypes`	Array of strings (Crimetypes) Optional crime type filter. Empty list means all crime types.

Responses

> **200** Forecast computed successfully.

— **400** Invalid startMonth/mode or invalid watchlist bbox configuration.

— **404** Watchlist not found for the authenticated user.

> **422** Validation Error

POST `/watchlists/{watchlist_id}/analytics/forecast`

Request samples

Payload

Content type
application/json

Copy Expand all Collapse all

```
{
  - "crimeTypes": [
        "Burglary",
        "Robbery"
    ],
    "mode": "drive",
    "startMonth": "2025-01"
}
```

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
    "generated_at": "2026-03-14T20:44:54Z",
  - "forecast": {
        "score": 100,
        "band": "red",
        "expected_crime_count": 56,
        "expected_collision_count": 0,
        "expected_collision_points": 0.4037,
      + "intervals": { … },
      + "components": { … }
    }
}
```


Read Root

Responses

> **200** Successful Response

GET /

Response samples

200

Content type
application/json

Copy

```
null
```

Health Check

Responses

> **200** Successful Response

GET /health

Response samples

200

Content type
application/json

Copy

```
null
```